

Homework 8

BSTA 550

Directions

Please turn in this homework on Sakai. Please submit your homework in pdf or html format.

You can type your work on your computer or submit a photo of your written work or any other method that can be turned into a pdf. The Adobe Scan phone app is an easy way to scan photos and compile into a PDF. Please let me know if you greatly prefer to submit a physical copy. We can work out another way for you to turn in homework.

You must show all of your work to receive credit.

Extra problems do not need to be turned in!

Questions

1. Let $X_i \sim \text{Binomial}(n_i, p)$ be independent r.v.'s for $i = 1, \dots, m$.
 - a. What does the r.v. $X = \sum_{i=1}^m X_i$ count, and what is the distribution of X ? Make sure to specify the parameters of X 's distribution.
 - b. Find $\mathbb{E}[X]$. *Make sure to show your work for (b) and (c). However, you may use without proof what you know about the mean and variance of each X_i .*
 - c. Find $\text{Var}[X]$.
2. Fill out the following table:

RV / Distribution	Type	Parameters	Possible values X	PMF / PDF	Mean $E[X]$	Variance $Var(X)$
Bernoulli	Discrete	p	$X = \{0, 1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial						
Geometric						
Negative Binomial						
Hypergeometric						
Poisson						
Discrete Uniform						
Continuous Uniform						
Exponential						
Gamma						
Normal						

3. In the following web, please fill in the relationships between each distribution:

